AI/NLP Risk Engine

Turning news & social text into financial risk signals, with a sentiment-driven index rebalancer and an event-driven stress tester

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Problem & Approach

Problem

Markets react to news and social posts faster than structured data arrives. Risk desks need explainable, machine-readable signals, not black boxes.

Approach

One pipeline outputs, per text: Sentiment (-1..1), Event class and Impact (1-10). Impact blends event severity, sentiment, source credibility, recency and novelty.

Ingest

News + tweets, entity/ticker extraction, de-duplication

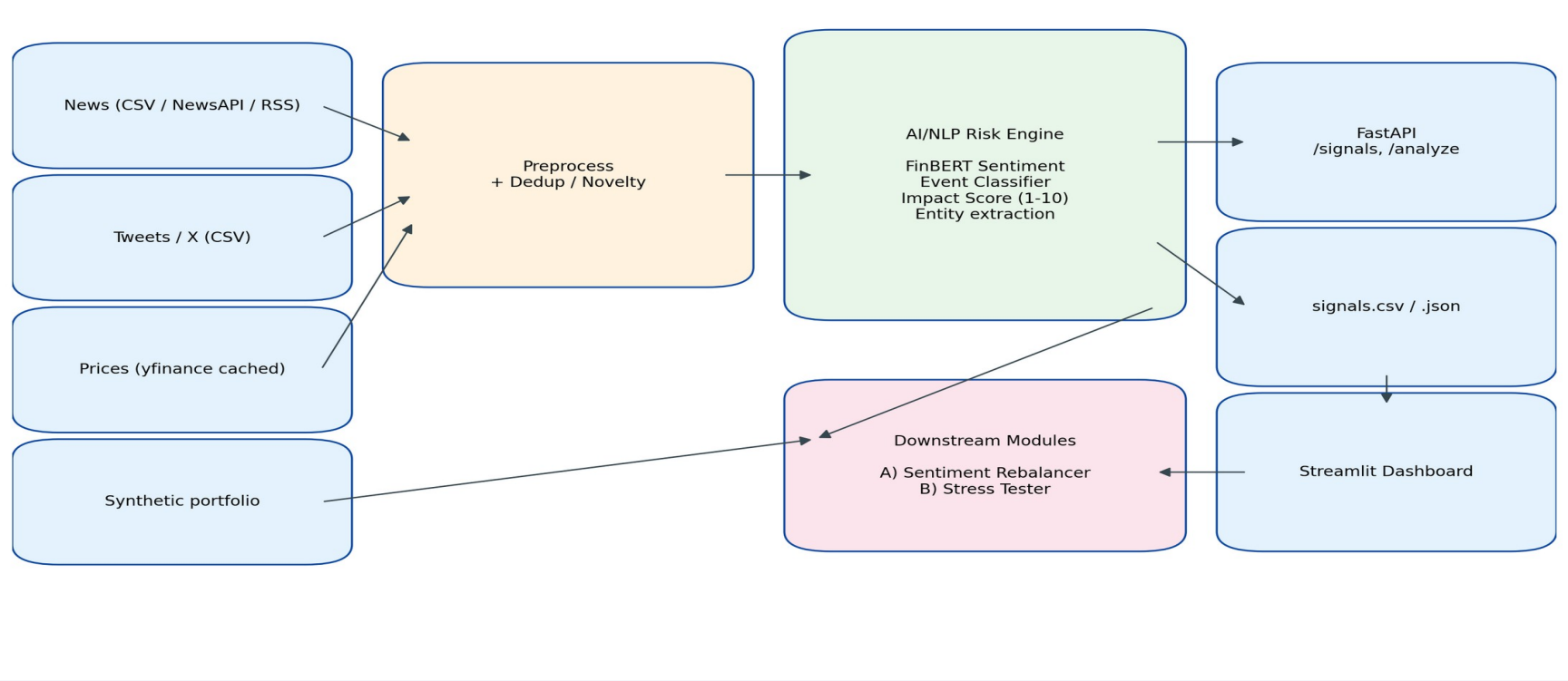
Understand

FinBERT / VADER sentiment, event classifier, impact score

Act

Module A rebalancer, Module B stress test, FastAPI + dashboard

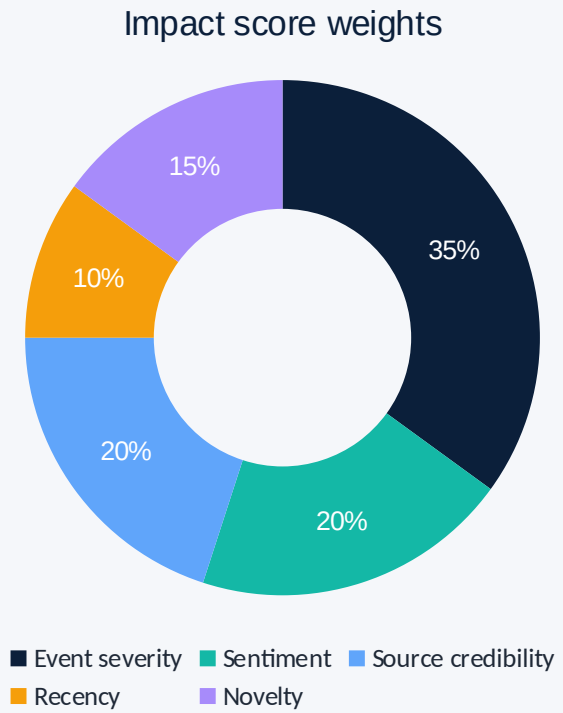
System Design



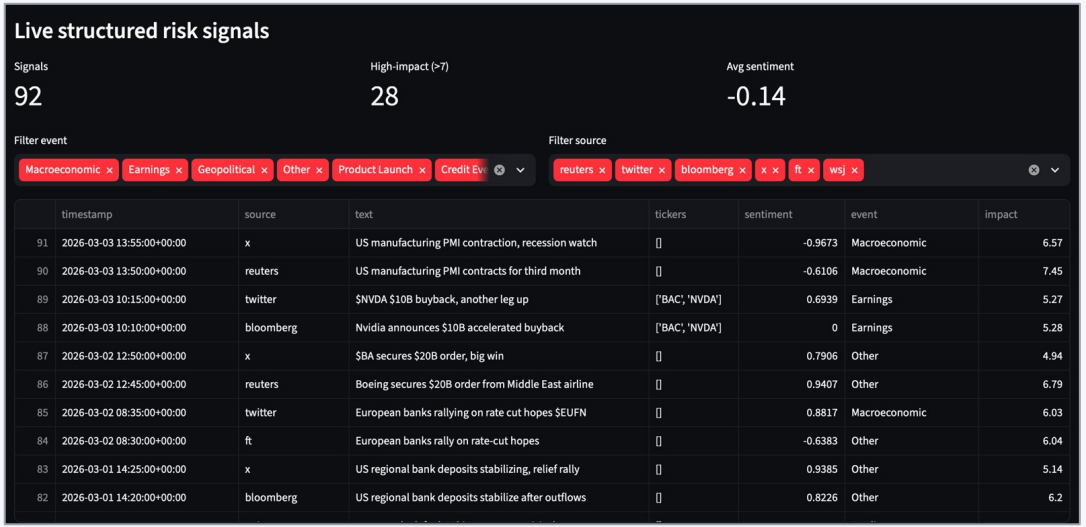
Python 3.11 | HuggingFace FinBERT | scikit-learn | FastAPI | Streamlit + Plotly | yfinance | pytest

Implementation Highlights

- Sentiment: FinBERT, with a VADER + finance-lexicon fallback for CPU/offline use
- Event classes: Geopolitical, Credit, Macro, M&A, Regulatory, Earnings, Product Launch, Operational (keyword baseline, optional zero-shot)
- Impact = 1 + 9 x weighted sum of 5 factors, so it is auditable
- Novelty via TF-IDF similarity stops duplicate stories inflating impact
- Module A: decayed, impact-weighted sentiment tilts weights, capped at 15%
- Module B: bond duration/convexity, loan PD/LGD, derivative delta/vega; auto-trigger when impact > 7

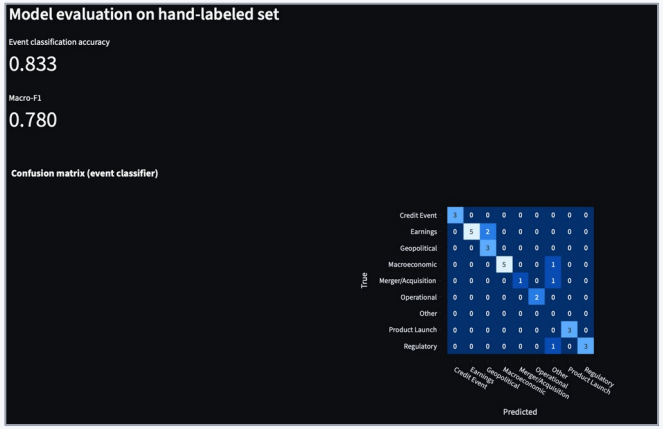


Key Results: NLP Risk Engine



	timestamp	source	text	tickers	sentiment	event	impact
91	2026-03-03 13:55:00+00:00	x	US manufacturing PMI contraction, recession watch	[]	-0.9673	Macroeconomic	6.57
90	2026-03-03 13:50:00+00:00	reuters	US manufacturing PMI contracts for third month	[]	-0.6106	Macroeconomic	7.45
89	2026-03-03 10:15:00+00:00	twitter	$NVDA $10B buyback, another leg up	['BAC', 'NVDA']	0.6939	Earnings	5.27
88	2026-03-03 10:10:00+00:00	bloomberg	Nvidia announces $10B accelerated buyback	['BAC', 'NVDA']	0	Earnings	5.28
87	2026-03-02 12:50:00+00:00	x	$BA secures $20B order, big win	[]	0.7906	Other	4.94
86	2026-03-02 12:45:00+00:00	reuters	Boeing secures $20B order from Middle East airline	[]	0.9407	Other	6.79
85	2026-03-02 08:35:00+00:00	twitter	European banks rallying on rate cut hopes $EUFN	[]	0.8817	Macroeconomic	6.03
84	2026-03-02 08:30:00+00:00	ft	European banks rally on rate-cut hopes	[]	-0.6383	Other	6.04
83	2026-03-01 14:25:00+00:00	x	US regional bank deposits stabilizing, relief rally	[]	0.9385	Other	5.14
82	2026-03-01 14:20:00+00:00	bloomberg	US regional bank deposits stabilize after outflows	[]	0.8226	Other	6.2

Live Signals tab: every text becomes a signal with sentiment, event class and impact score



Event classifier on 30 hand-labelled headlines (small set, so indicative only)

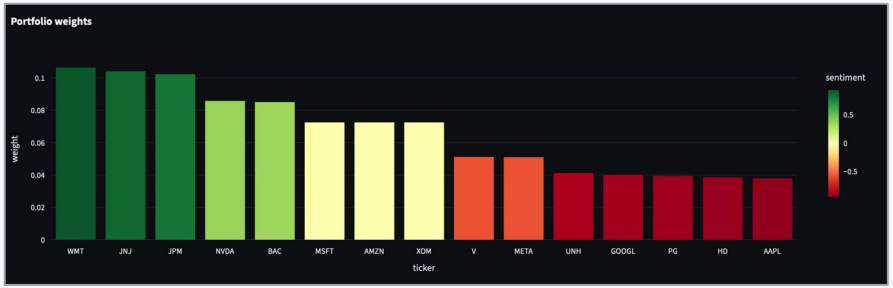
92
signals processed

28
high-impact (>7)

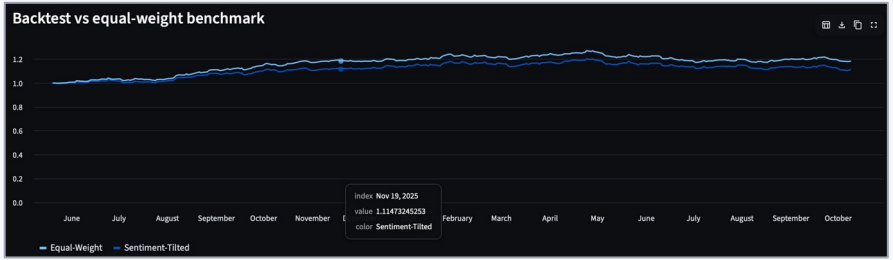
0.833
event accuracy

0.780
event macro-F1

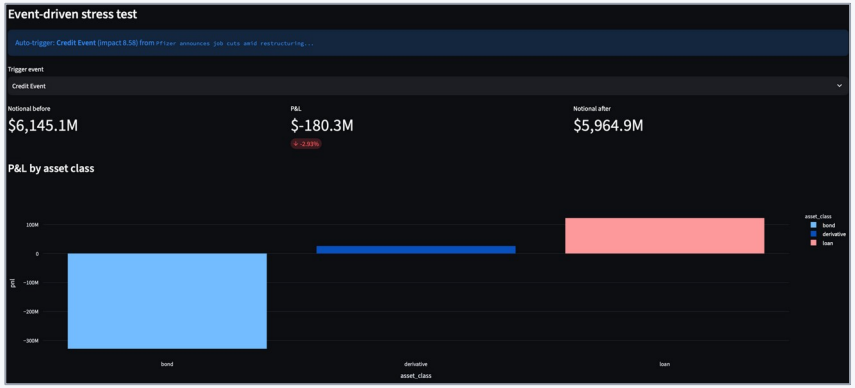
Key Results: Modules A & B



Module A: WMT, JNJ, JPM overweight; AAPL, HD, PG underweight



Backtest, constant weights: cum. return 11.0% vs 18.4% equal-weight; Sharpe 1.03 vs 1.73; max drawdown -8.2% vs -7.7%



Module B: Credit Event auto-trigger (impact 8.58) gives P&L of -$180.3M (-2.93%) on $6.1B notional

- Tilted index trailed equal-weight in this test; weights from Feb-Mar 2026 signals are applied to a different price window, so no alpha is claimed
- Stress test shows P&L by asset class and the top-10 affected positions; bonds drive the largest loss

Domain Impact, Limitations & Next Steps

Wholesale banking risk

Credit and macro events flow into loan, bond and derivative exposures.

Explainable and auditable

Each impact score traces to event prior, source credibility, recency and novelty.

Reusable pipeline

Same API and module interface for ratings surveillance or index operations.

Limitations

- Synthetic data; no live ingestion yet
- Rule-based classifier, hand-tuned impact weights
- Backtest holds weights constant and trailed equal-weight
- Alias-based ticker matching can mis-tag companies

Next steps

- Day-by-day replay rebalancer with costs
- Fine-tuned classifier; calibrate impact on realized moves
- Live NewsAPI / GDELT ingestion
- Correlation-aware stress tests